

WELCOME TO PROJECT NEW HORIZON

SHARED PROJECT

TEAM ID	CONTINENT	NAMES
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JCEC is studying whether a small human research district could be built safely on your assigned continent. Your team will design a three-building prototype district, test one important structure, improve it using evidence, and decide whether the structural proposal should be authorized.

Important: Phase 4 does not decide whether humans *should* settle Jabberwocky. The final Mission 2190 Council makes that decision later.

THE MAIN CHALLENGE

Can your team design, test and defend a safe, efficient JCEC research district that could be built if the final Council later approves human presence?

YOUR DISTRICT MUST INCLUDE

- ☐ 1 circular-base building
- ☐ 1 triangular-base building
- ☐ 1 parallelogram-base building
- ☐ 1 Primary Structural Test Building
- ☐ a measured district blueprint
- ☐ required road constructions
- ☐ one simple energy plan
- ☐ one simple transportation plan
- ☐ established environmental/site restrictions

ONE PRIMARY STRUCTURAL TEST BUILDING

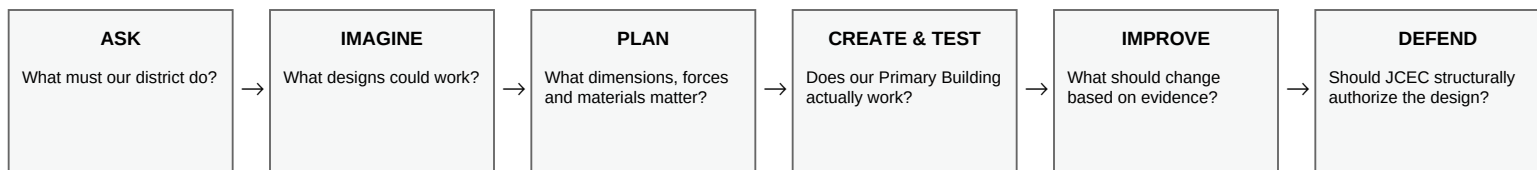
Only **one** of the three buildings completes the full Science testing cycle:

requirements → force map → material/joint testing → graph → stability test → weakness → two redesigns → modify → retest → authorization

Simple + evidence-based beats large + complicated.

Decoration is optional and ungraded.

ENGINEERING DESIGN PROCESS



THREE EVIDENCE LANES

SCIENCE	MATH	SHARED PROJECT
Requirements → Forces → Materials → Stability → Authorization	Building Geometry → Geometric Constructions → Resource Mathematics	Blueprint → Model → Energy / Transportation → Final District

PAGE FINISH LINE

We understand what we are building, what counts as evidence, and that appearance is not the goal.

MISSION 1 · JCEC DISTRICT DESIGN BRIEF

SCIENCE EVIDENCE

TEAM ID CONTINENT NAMES

TEAM RECORD · NEW HORIZON REQUIREMENTS CARD

Use the inherited briefing. Do not reopen old mission archives.

Inherited finding	Evidence / note
ENVIRONMENT	
THERMAL REQUIREMENT	
ECOLOGICAL RESTRICTION	
SITE / LOAD WARNING	

OUR DISTRICT'S JOB

PURPOSE: Small long-term JCEC research district prototype

What must this district successfully allow JCEC researchers to do?

THREE PERFORMANCE REQUIREMENTS

A performance requirement describes something the structure must be able to do safely and reliably. Do not name your final structural solution yet.

Requirement	Why does the continent make this important?
1. Our district must...	
2. Our district must...	
3. Our district must...	

STRUCTURAL THINKING

Frame Shell Frame + shell Not enough evidence yet

One likely failure point we need to investigate:

PRIMARY STRUCTURAL TEST BUILDING · TENTATIVE CHOICE

Circular building Triangular building Parallelogram building

Why might this be an important structure to test?

INDIVIDUAL REFLECTION Why can two structures with the same purpose have very different designs?

FINISH LINE: 3 requirements · 1 restriction · 1 possible failure point · tentative Primary Building

MATH EVIDENCE 1 · BUILDING GEOMETRY

MATH EVIDENCE

NAME TEAM CONTINENT

Your team may choose the same building dimensions, but you must complete your own calculations. Show your work. Use appropriate units.

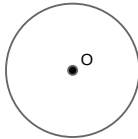
A. CIRCULAR BUILDING · GRADE 7 MATH EVIDENCE

Radius = Diameter = How are radius and diameter related?

Calculation	Show your work	Answer
Circumference		cm
$C \div d \approx$		
Area		cm ²

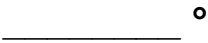
CONSTRUCT THE CIRCULAR FOOTPRINT

Mark centre O. Use the required radius to construct the circle. Label one radius and one diameter.



CENTRAL ANGLES

The angles that meet at the centre of a complete circle total:



B. TRIANGULAR BUILDING

Base = Perpendicular height = AREA — GRADE 7 MATH EVIDENCE

Formula: Calculation: Area:

PERIMETER — REVIEW / APPLICATION Side lengths: Perimeter:

C. PARALLELOGRAM BUILDING

Base = Perpendicular height = AREA — GRADE 7 MATH EVIDENCE

Formula: Calculation: Area:

PERIMETER — REVIEW / APPLICATION Perimeter:

MATH EVIDENCE CHECK

circle relationships circumference + π circle area circle construction 360° central angles triangle area parallelogram area

IMAGINE · DISTRICT CONCEPT PLAN

SHARED PROJECT

TEAM ID CONTINENT NAMES

Sketch **at least two possible district arrangements** before choosing one. These are rough ideas. Do not worry about perfect measurements yet.

IDEA A

Label: circular building · triangular building · parallelogram building · Primary Test Building · main road/path · protected/restricted area

IDEA B

Label: circular building · triangular building · parallelogram building · Primary Test Building · main road/path · protected/restricted area

COMPARE YOUR IDEAS

Idea A advantage:

Idea A concern:

Idea B advantage:

Idea B concern:

TEAM DECISION

We will develop Idea because:

LIGHT PLANNING ONLY

ENERGY PLAN — one idea:

TRANSPORTATION PLAN — one idea:

You are **not** completing an energy or transportation research project.

PAGE FINISH LINE

We considered at least two arrangements and selected one for evidence-based reasons.

PLAN · MEASURED DISTRICT BLUEPRINT

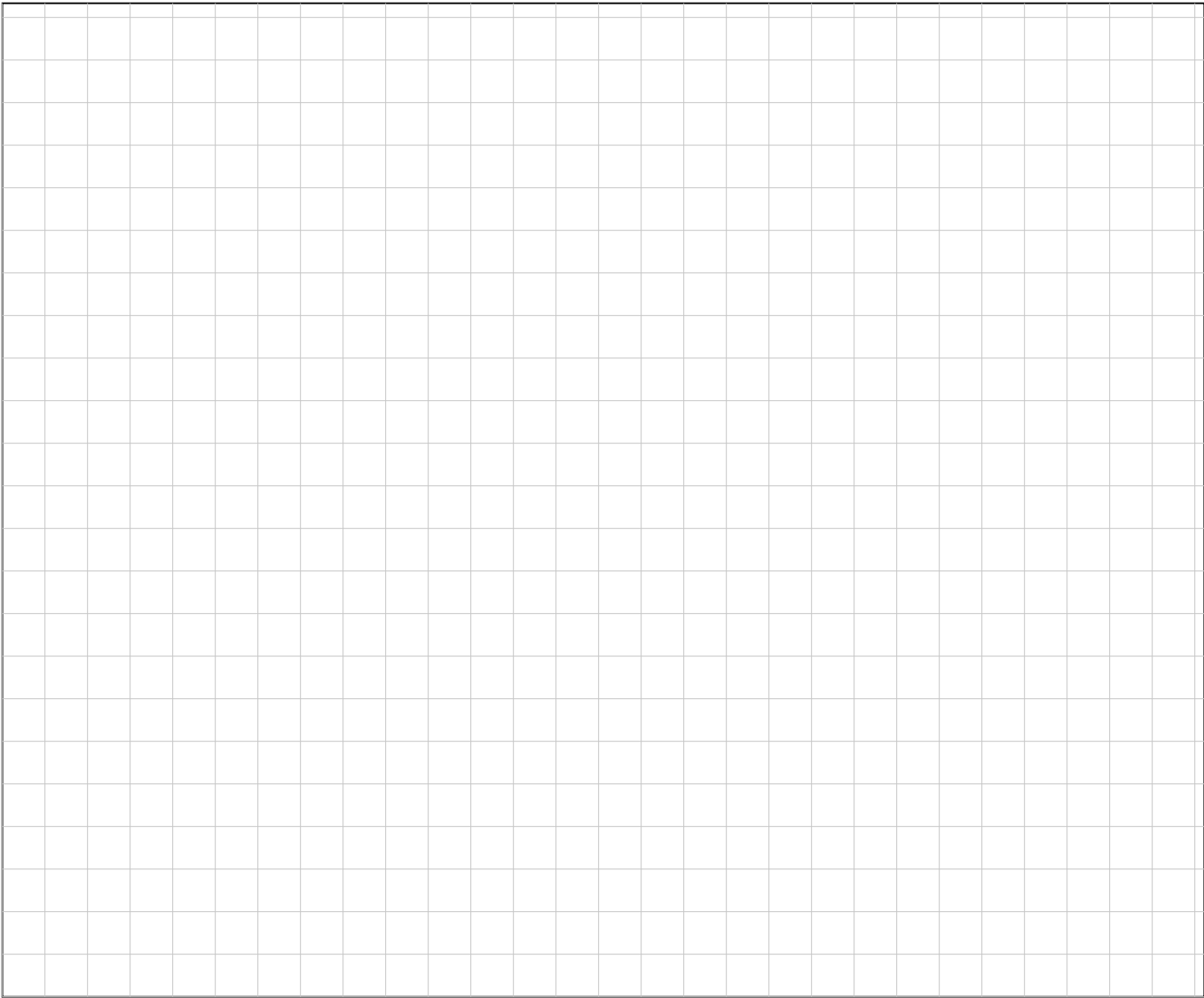
SHARED PROJECT

TEAM ID CONTINENT NAMES

THIS IS A MEASURED MODEL PLAN — NOT A FORMAL SCALE-FACTOR ASSIGNMENT.
Use the grid to organize your district. Write the actual planned model dimensions beside important features. You do not need to calculate real-world city dimensions.

BLUEPRINT REQUIREMENTS

- ☐ circular footprint
- ☐ triangular footprint
- ☐ parallelogram footprint
- ☐ Primary Test Building marked
- ☐ actual model dimensions
- ☐ roads / paths
- ☐ protected or restricted area
- ☐ simple energy symbol
- ☐ transportation route



BLUEPRINT-TO-MODEL CHECK

Planned model-base length:Planned model-base width:

Largest building footprint:One site restriction visible:

MATH EVIDENCE 2 · GEOMETRIC CONSTRUCTIONS

MATH EVIDENCE

NAME _____ TEAM _____ CONTINENT _____

Construction marks matter. Do not simply sketch lines that look parallel or perpendicular. Show the construction method your teacher taught you.

A. CONSTRUCT PARALLEL LINES

How do you know the lines are parallel?

Where could this appear in our district?

B. CONSTRUCT PERPENDICULAR LINES

How do you know the lines are perpendicular?

District application:

C. CONSTRUCT A PERPENDICULAR BISECTOR

Verify: two equal parts at 90°.

How did you verify both conditions?

D. CONSTRUCT AN ANGLE BISECTOR

Divide one angle into two equal angles.

Original angle: ____° New angles: ____° and ____°

BLUEPRINT CONNECTION

Circle where these appear on your team's plan: ■ parallel ■ perpendicular ■ perpendicular bisector ■ angle bisector

MISSION 2 · STRUCTURAL FORCE MAP

SCIENCE EVIDENCE

TEAM ID CONTINENT NAMES

Draw the **structural system**, not the decoration. Show the forces that matter for your Primary Structural Test Building.

PRIMARY STRUCTURAL TEST BUILDING · FORCE MAP

EXTERNAL LOAD ↓ / → / ← TENSION = pulling COMPRESSION = squeezing SHEARING = sliding BENDING = curving FRICTION = resisting sliding

REQUIREMENTS ON YOUR DIAGRAM:

■ at least one important external load ■ force direction arrows ■ internal-force labels where evidence supports them ■ one likely failure point

MASS VS FORCE MINI-CHECK

Mass tells us Mass units

Force tells us Force unit

ONE TEAM DECISION

Our priority load / force risk is:

because:

TEAM RECORD · MISSION 2 — STRUCTURAL FORCE MAP

Individual reflection: How can one external load create different internal forces in different parts of a structure?

MISSION 3 · MATERIAL & JOINT FAIR TEST

SCIENCE EVIDENCE

TEAM ID _____ CONTINENT _____ NAMES _____

Investigation question: Which material and joint behaviour best fit the job of our Primary Structural Test Building?

A. OPERATIONAL DEFINITIONS

STIFFNESS _____

FLEXIBILITY _____

STRENGTH _____

Do not test materials to dangerous destruction. Stop at the teacher-set maximum.

B. FAIR TEST

WHAT WILL WE CHANGE? _____

WHAT WILL WE MEASURE? _____

WHAT MUST STAY SIMILAR?

1. _____
2. _____
3. _____

C. LOAD-DEFORMATION DATA

Load step / safe load	Material A deformation (mm)	Material B deformation (mm)	Material C deformation (mm)
1			
2			
3			
4			
Teacher maximum			

Material A: _____ Material B: _____ Material C: _____

D. JOINT CHECK

Joint	What happened?	Best use / limitation
Fixed joint		
Flexible joint		

FINISH LINE: We have quantitative material evidence and joint evidence ready to graph and interpret.

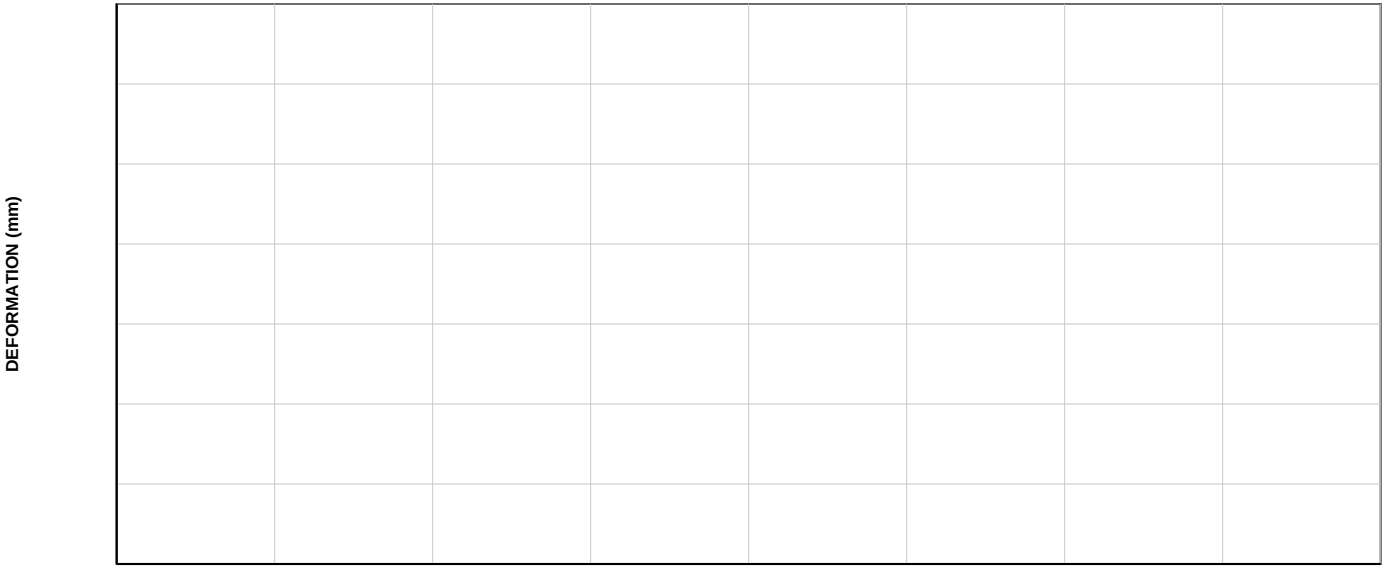
MISSION 3 · LOAD–DEFORMATION GRAPH + MATERIAL DECISION

SCIENCE EVIDENCE

TEAM ID CONTINENT NAMES

A. LOAD–DEFORMATION GRAPH

Plot at least two candidate materials on the same graph. Use a clear legend. x-axis = load / load step. y-axis = deformation (mm).



Material 1: LOAD / LOAD STEP Material 2:

B. WHAT DOES THE EVIDENCE SHOW?

Which material was stiffer under comparable loads?

Which material showed useful flexibility?

What evidence do you have about strength?

Safeguard: Reaching the teacher-set maximum does not prove the exact breaking strength.

C. MATERIAL & JOINT RECOMMENDATION

For part of our Primary Building, we recommend material:

Joint:

Our strongest numerical evidence is:

TEAM RECORD · MISSION 3 — MATERIAL & JOINT RECOMMENDATION

Individual reflection: Why isn't the material that holds the greatest load automatically the best material for every part of a structure?

One thing our test does NOT prove is:

MATH EVIDENCE 3 · JCEC RESOURCE BUDGET

MATH EVIDENCE

NAME TEAM CONTINENT

JCEC purchases materials in standard resource units. You do not need to measure every centimetre of tape you touch. Starting budget: \$200.00.

PRICE LIST

Table with 2 columns: Resource unit, Cost. Rows include 1 sheet cardstock, 30 x 30 cm cardboard, 5 straws, 5 craft sticks, 50 cm masking tape, 10 mL white glue, 50 cm string.

A. PLANNED BUDGET

Table with 4 columns: Material, Qty, Unit cost, Planned cost. Includes a row for Planned total =

B. ACTUAL BUILD + REDESIGN

Table with 4 columns: Material, Qty actually used, Unit cost, Actual cost. Includes a row for Actual build total = and a row for Redesign cost =

C. INDIVIDUAL CALCULATIONS

Total project cost:
Money remaining: \$200.00 - \$ = \$
Percentage of budget used: project cost ÷ 200 × 100 = %
OPTIONAL / TEACHER-SELECTED: Average cost per main building: \$ ÷ 3 = \$

BUDGET REASONING

Did your team spend more than planned? Why or why not?

FINISH LINE: I can use decimal operations and percent to explain our project budget.

MISSION 4 · PRIMARY BUILDING STABILITY TEST

SCIENCE EVIDENCE

TEAM ID _____ CONTINENT _____ NAMES _____

Investigation question: What makes our Primary Structural Test Building stable, and where is it vulnerable?

A. BEFORE TESTING

We think the building's most important stability weakness will be:

WHY?

☐ foundation / support

☐ mass distribution

☐ shape

☐ joint / fastening

☐ bending

☐ other

B. KEEP THE TEST FAIR

Teacher-selected test: _____

What must stay the same when we retest?

1. _____
2. _____
3. _____

C. BASELINE TEST EVIDENCE

Trial	Test condition	Quantitative result, if available	What happened?
1			
2			
3			

D. OBSERVED WEAKNESS

The most important weakness was: _____

Evidence: _____

E. STABILITY SCIENCE

Foundation / support effect: _____

Mass distribution effect: _____

One strengthening method that could help: _____

FINISH LINE: We have enough baseline evidence to propose more than one improvement.

MISSION 4 · ALTERNATIVES → REDESIGN → RETEST

SCIENCE EVIDENCE / SHARED PROJECT

TEAM ID _____ CONTINENT _____ NAMES _____

Do not jump straight to your favourite fix. Propose two reasonable alternatives first.

ALTERNATIVE A

Change: _____

Science reason: _____

Possible disadvantage: _____

Estimated extra resource cost: \$ _____

ALTERNATIVE B

Change: _____

Science reason: _____

Possible disadvantage: _____

Estimated extra resource cost: \$ _____

C. CHOOSE

We selected A / B because: _____

D. MODIFY

What exactly changed on the model? _____

What stayed the same so the retest is useful? _____

E. RETEST

Evidence	Baseline	Redesigned
Main quantitative result		
Main observation		
Failure / weak point		

RESULT: ☐ Improved ☐ About the same ☐ Worse ☐ Unclear

A worse result is still evidence. What does the retest tell us?

F. NEW PROBLEM OR LIMITATION

Did the modification solve one weakness but create another?

FINISH LINE: We tested an evidence-based redesign using a comparable procedure.

MISSION 4 · NEW HORIZON SAFETY PROTOCOL + FINAL BUILD READINES

SCIENCE EVIDENCE

TEAM ID	CONTINENT	NAMES
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TEAM RECORD · NEW HORIZON SAFETY PROTOCOL

STABILITY RULE · JCEC should require:

FOUNDATION / MASS RULE:

STRENGTHENING RULE:

ENVIRONMENTAL / SITE CONDITION:

MARGIN OF SAFETY

A margin of safety means we do not design something to operate right at the edge of failure.

Our rule:

ONE IMPORTANT UNCERTAINTY

FINAL CITYSCAPE READINESS CHECK

THREE-BUILDING DISTRICT ☐ Circular-base building complete ☐ Triangular-base building complete ☐ Parallelogram-base building complete ☐ Primary Test Building identified	EVIDENCE ☐ Model reflects M1 requirements ☐ Primary Building responds to M2 Force Map ☐ Material choice matches M3 evidence ☐ M4 redesign visible or documented	DISTRICT PLAN ☐ Model reasonably matches blueprint ☐ Protected/restricted area respected ☐ Simple energy plan shown ☐ Simple transportation plan shown
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Decoration is optional and ungraded. Your district is ready when the Science evidence, Math evidence and functional model are complete.

TEAM RECORD · MISSION 4 — NEW HORIZON SAFETY PROTOCOL

Individual reflection: Why can adding more material make a structure less efficient—or even create a new problem?

MISSION 5 · JCEC NEW HORIZON STRUCTURAL AUTHORIZATION

SCIENCE EVIDENCE

TEAM ID CONTINENT NAMES

THIS IS A STRUCTURAL AUTHORIZATION. You are deciding whether this prototype works structurally. You are NOT deciding whether humans should settle Jabberwocky.

A. STRUCTURAL EVIDENCE LOCKER

Evidence source	Strongest evidence to carry forward
M1 · REQUIREMENTS	
M2 · FORCES	
M3 · MATERIALS	
M4 · SAFETY	

B. JCEC DESIGN REVIEW

Review lens	Evidence strong enough?	Concern / required revision
FUNCTION	■ Yes ■ Not yet	
RELIABILITY	■ Yes ■ Not yet	
SAFETY	■ Yes ■ Not yet	
MATERIAL EFFICIENCY	■ Yes ■ Not yet	
COST / COMPLEXITY	■ Yes ■ Not yet	
ENVIRONMENTAL IMPACT	■ Yes ■ Not yet	

C. FINAL STRUCTURAL AUTHORIZATION

☐ PROCEED☐ PROCEED AFTER REQUIRED MODIFICATIONS☐ LIMIT THE BUILD☐ HOLD FOR MORE EVIDENCE

Required modification / restriction / next evidence:

D. FINAL JCEC MESSAGE · 3–5 SENTENCES

Explain your authorization using evidence from at least three different missions.

TEAM RECORD · MISSION 5 — JCEC NEW HORIZON STRUCTURAL AUTHORIZATION

Phase 4 is finished when your team has one Structural Authorization and each student has completed the individual reflection. Structural feasibility has been investigated. The final Mission 2190 Council will later decide whether human presence should actually proceed.